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Part A.

Circuit in series

1. Total resistance in ohms

Total resistance (R) = 20 + 30 + 50

= 100 Ω

2. Total current from power supply

$I = V/R$

V = 125

R = 100

$\frac{125}{100} = 1.25$ A

3. Current through each resistor

since it is series its the same

I_1 = 1.25 A

I_2 = 1.25 A

I_3 = 1.25 A

4. Voltage drop across each resistor.

V = IR

20 Ω resistor = 1.25 X 20 = 25 V

30 Ω resistor = 1.25 x 30 = 37.5 V

50 Ω resistor = 1.25 x 50 = 62.5 V

5. Power Desipated through each resistor

Watts law : Power = Voltage * Current

20 Ω resistor = 25 X 1.25 = 31.25 W

30 Ω resistor = 37.5 x 1.25 = 46.875 W

50 Ω resistor = 62.5 x 1.25 = 78.125 W

Circuit in parallel.

1. Total resistance

$1/R_T = 1/20 + 1/100 + 1/50 = \frac{5+1+2}{100}$

$= 8/100$

$1/R_T = 0.08$

R_T = inverse of 0.08

$= 1/0.08$

$R_T = 12.5\ \Omega$

2. Total current from power supply

ohms law:

$I = V/R = 125/12.5$

$I = 10A$

3. Current through each resistor

in parallel voltage is same across:

$20\ \Omega$ resistor $= 125/20 = 6.25\ A$

$100\ \Omega$ resistor $= 125/100 = 1.25\ A$

$50\ \Omega$ resistor $= 125/50 = 2.5\ A$

4. Voltage drop across each resistor

$V_1 = 125V$

$V_2 = 125V$

$V_3 = 125V$

5. Power dissipated through each resistor:

$P = VI$

$20\ \Omega$ resistor : $P = 125 \times 6.25 = 781.25W$

$100\ \Omega$ resistor : $P = 125 \times 1.25 = 156.25W$

$50\ \Omega$ resistor : $P = 125 \times 2.5 = 312.5W$